

To Speak, See, and Think - Evolution of Large Language Models in Healthcare

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Associate Director of AI, Preston A. Wells Jr. Center for Brain Tumor Therapy

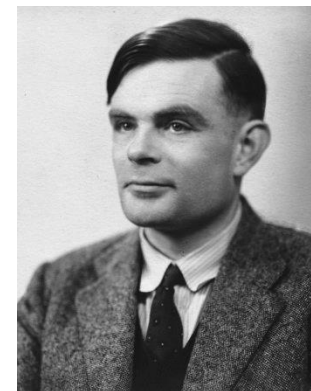
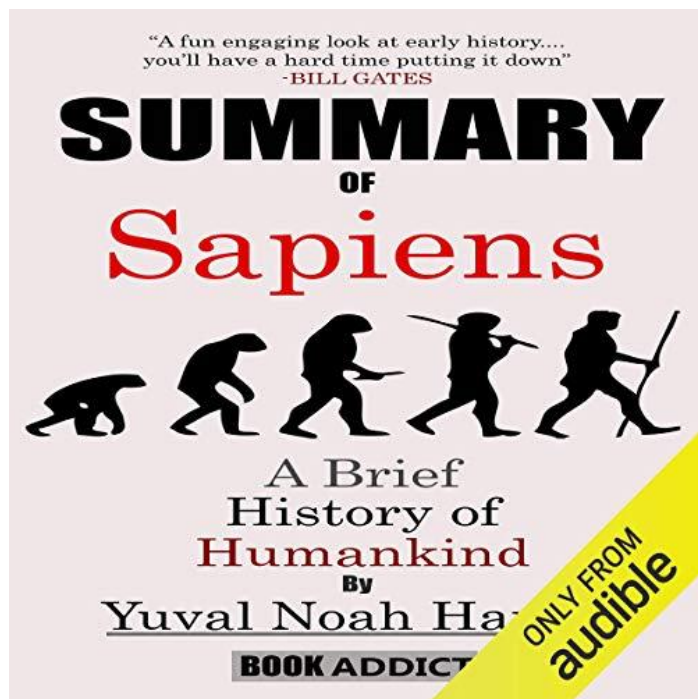
Chief Data Scientist, Director of Natural Language Processing



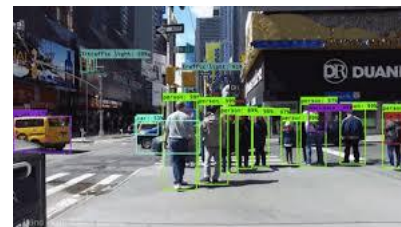
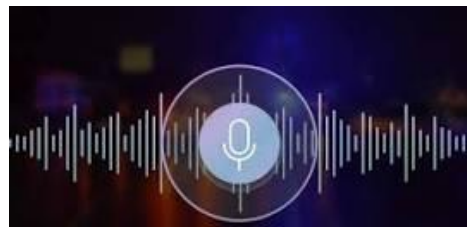
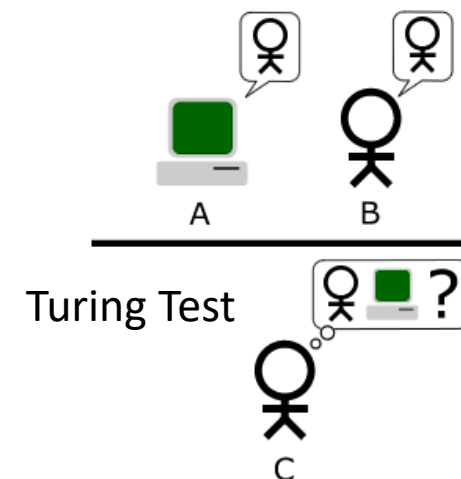
Disclosure

- Competing Interests: None
- Funding support
 - **PCORI ME-2023C3-35934 (PI, contact)** - Transfer learning NLP to improve adoption of clinical text in multi-site studies.
 - **PCORI ME-2018C3-14754 (PI, contact)** - NLP to Connect Social Determinants and Clinical Factors for Outcomes Research.
 - **ARPA-H PARADIGM (MPI)** - Multimodal Multi-Medical Task Adaptive Guidance System (Multi-TAGs).
 - **NIH/NIA R56AG069880 (PI, contact)** - Advancing Drug Repositioning for Alzheimer's Disease using Real-word data.
 - **NIH/NCATS UF CTSA (Chief Data Scientist)** - Together: Transforming and Translating Discovery to Improve Health.
 - **NIH/NIA U24AG098157 (UF Site PI)** - ReCARDO: Using Real-World Data to Derive Common Data Elements for Alzheimer's Disease and AD-Related Dementias Research Through Ontological Innovation
 - NIH/NHLBI R01HL176844 - AI-Enabled COMPLETE Model for Better Prediction of CLTI Revascularization Outcomes
 - NIH/NIDA R01DA063631 - Prediction of Relapse on OUD Treatment using Machine Learning-Driven Evidence-based Clinical Decision Support Tool (PROTECT)
 - NIH/NIMH R01MH121907 - Linking VA and non-VA data to study the risk of suicide in chronic pain patients.
 - NIH/NIDA R01DA057886 Assessing Performance of a Hepatitis C Emergency Department (HepC-EnD) Screening Tool.
 - NIH/NIAID R01AI172875 - Artificial Intelligence and counterfactually actionable response to end HIV (AI-CARE HIV).
 - NIH/NHLBI R01HL169277 - Identifying pediatric asthma subtypes using novel privacy-preserving federated machine learning methods.
 - NIH/NCI R01CA284646 - Advancing Precision Lung Cancer Surveillance and Outcomes in Diverse Populations (PLuS2).
 - NIH/NCI R37CA272473 - De-implementation of inappropriate thyroid ultrasound.
 - NIH/NIDA R01DA050676 - Developing and Evaluating a Machine-Learning Opioid Prediction and Risk-Stratification E-Platform (DEMONSTRATE).
 - CDC U18DP006512 - Using Real-world Data to Assess the Burden of Diabetes in Children and Adolescents.

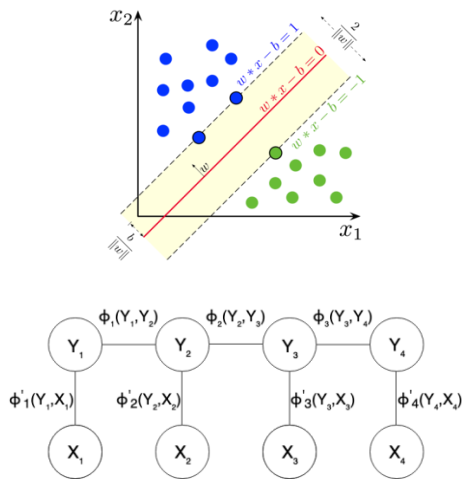
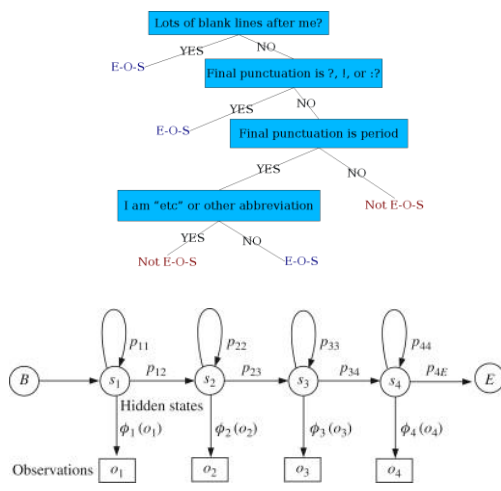
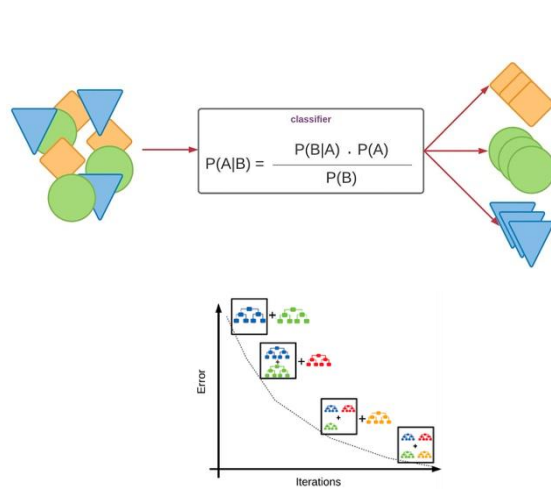
Artificial Intelligence



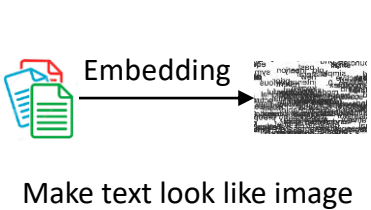
Alan Turing (1912-1954)
British mathematician, logician, and computer scientist



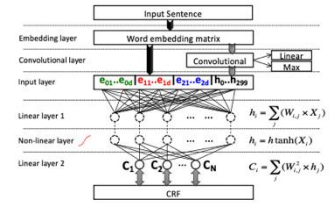
Learn to be intelligent



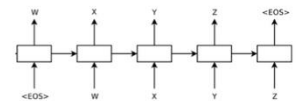
NLP from scratch



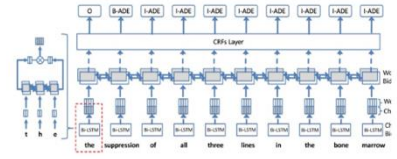
Make text look like image



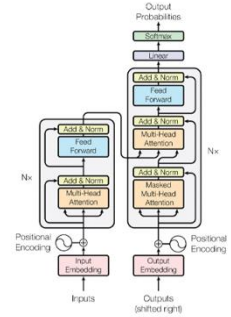
Sequence-to-sequence learning



LSTM-CRFs



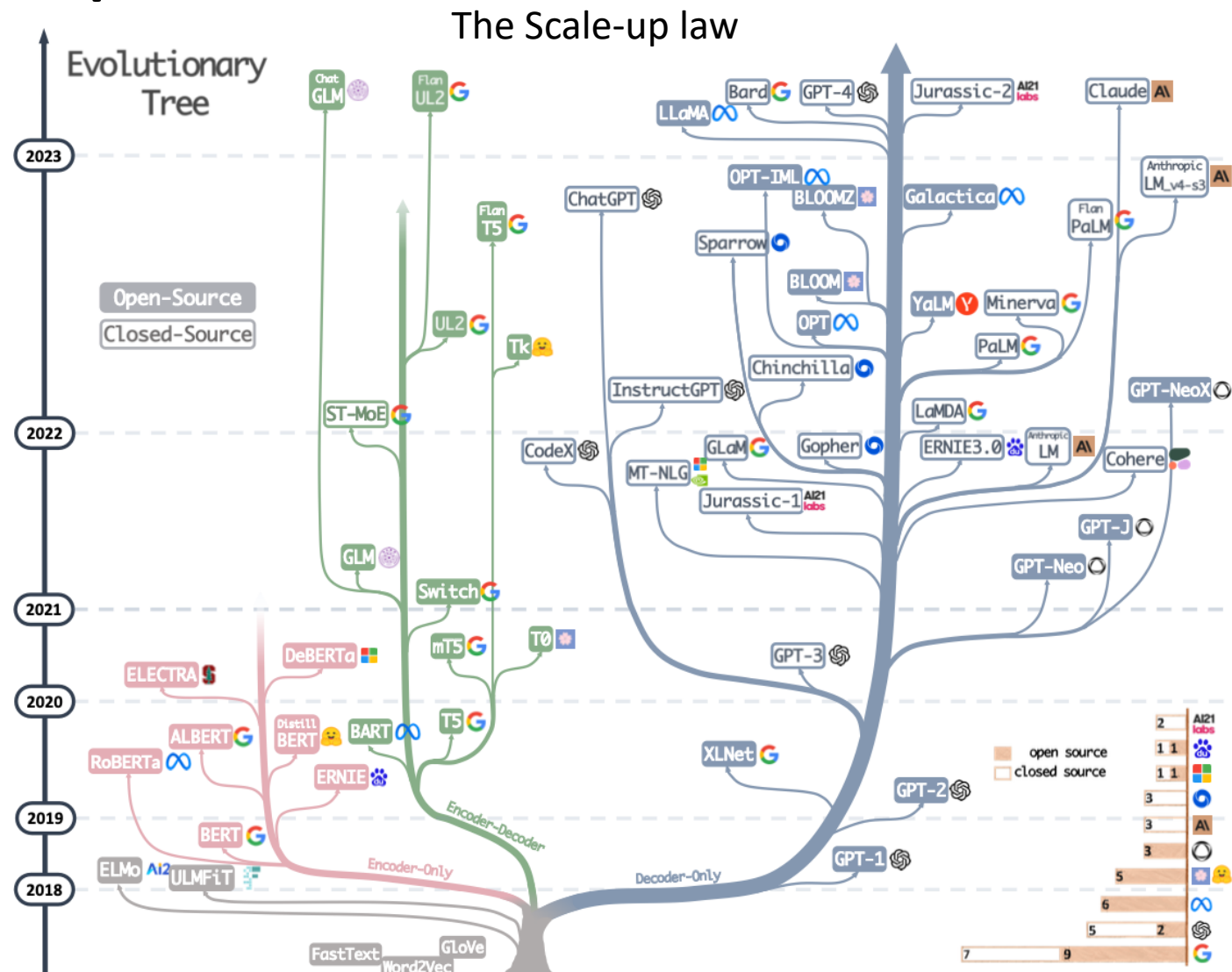
Transformer



ChatGPT



Learn to speak



Biomedical LLMs

- Medical LLMs
 - Pre-training from scratch
 - Post-training (self-supervised continuous training, supervised fine-tuning)
 - Instruction training using RL (Reinforcement learning)
- ClinicalBERT
 - *1.2 billion words from 2 million clinical notes*
- BioBERT
 - *21 billion words of text: 2.5 billion words from English Wikipedia, 0.8 billion words from the BooksCorpus, 4.5 billion words from PubMed abstracts, and 13.5 billion words from PMC full-text articles*
- PubMedBERT
 - Trained from scratch using PubMed literature

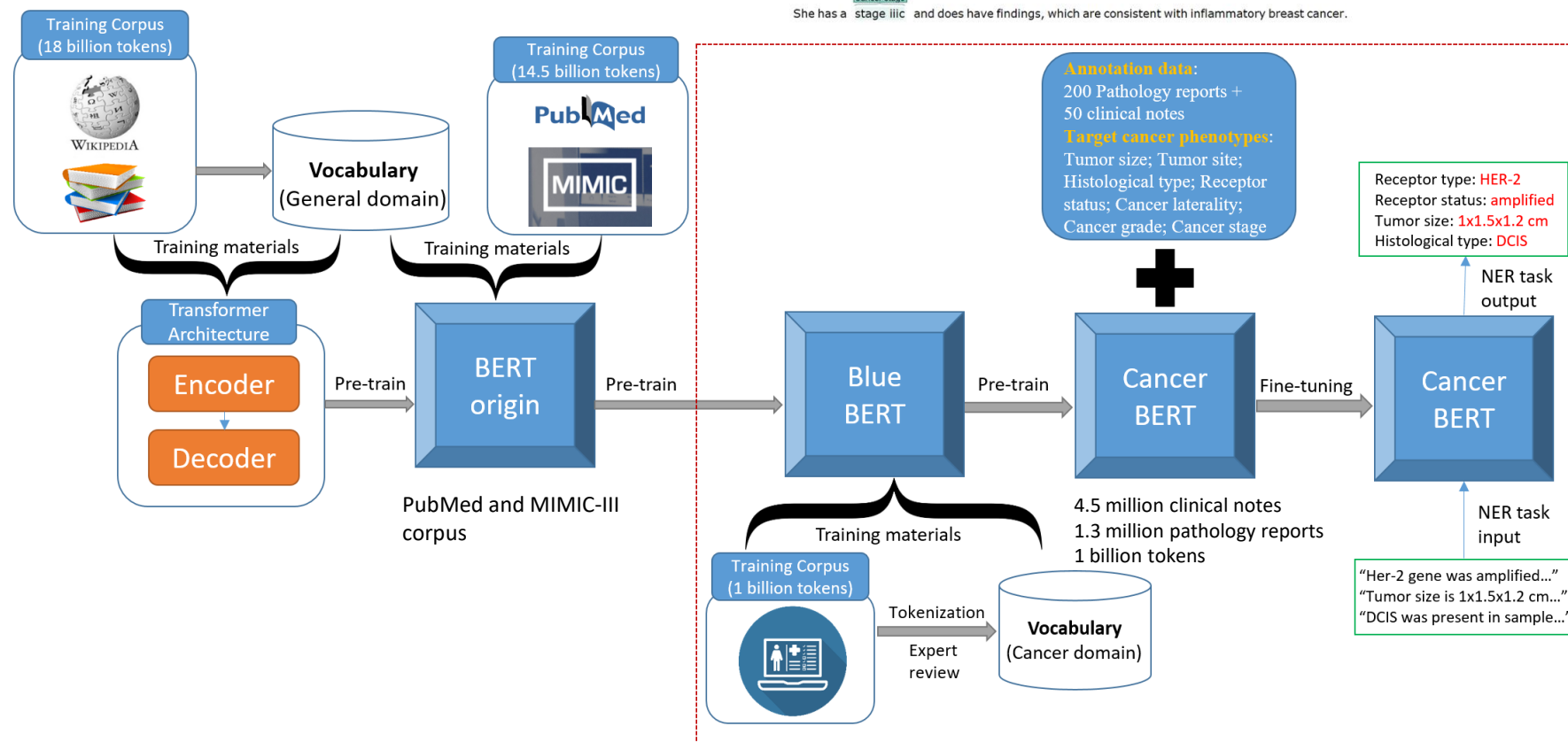
A cancer domain-specific language model

CancerBERT: a cancer domain-specific language model for extracting breast cancer phenotypes from electronic health records

Sicheng Zhou, Nan Wang, Liwei Wang, Hongfang Liu, Rui Zhang

Journal of the American Medical Informatics Association, Volume 29, Issue 7, July 2022, Pages 1208–1216, <https://doi.org/10.1093/jamia/ocac040>

Annotation
Ultrasound-guided core biopsies: Infiltrating lobular carcinoma (Nottingham Grade 2 of 3, score 6 of 9), (E-Cadherin stain performed and negative). Focal lobular carcinoma in situ present, no evidence of lymphovascular space involvement. Estrogen receptor and progesterone receptor studies completed and both positive. 50 u/l ast 17 0-45 u/l final history of present illness: clauden cochrum is in today for follow-up of her left-sided infiltrating ductal carcinoma with apocrine features diagnosed 04/28/2009. She had a lumpectomy on 05/15/2010 for a 1.4 cm, grade 2 of 3 cancer with 0 of 3 lymph nodes involved. Impression and plan: Ms.xxx is a pleasant 52-year-old woman with a history of newly diagnosed infiltrating ductal carcinoma of the left breast cancer. She is a clinical t4 n2 mx with evidence of left axillary adenopathy. She has a stage iiic and does have findings, which are consistent with inflammatory breast cancer.



NYUTron

- Readmission
- Mortality
- Comorbidity
- Length of stay
- Insurance denial

Article

Health system-scale language models are all-purpose prediction engines

<https://doi.org/10.1038/s41586-023-06160-y>

Received: 14 October 2022

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Published online: 07 June 2023

Open access

 Check for updates

Lavender Yao Jiang^{1,2}, Xujin Chris Liu^{1,3}, Nima Pour Nejatian⁴, Mustafa Nasir-Moin¹, Duo Wang⁵, Anas Abidin⁴, Kevin Eaton⁶, Howard Antony Riina¹, Ilya Laufer¹, Paawan Punjabi⁶, Madeline Miceli⁶, Nora C. Kim¹, Cordelia Orillac¹, Zane Schnurman¹, Christopher Livia¹, Hannah Weiss¹, David Kurland¹, Sean Neifert¹, Yosef Dastagirzada¹, Douglas Kondziolka¹, Alexander T. M. Cheung¹, Grace Yang^{1,2}, Ming Cao^{1,2}, Mona Flores⁴, Anthony B. Costa⁴, Yindalon Aphinyanaphongs^{5,7}, Kyunghyun Cho^{2,8,9,10} & Eric Karl Oermann^{1,2,11}✉

Physicians make critical time-constrained decisions every day. Clinical predictive models can help physicians and administrators make decisions by forecasting clinical and operational events. Existing structured data-based clinical predictive models have limited use in everyday practice owing to complexity in data processing, as well as model development and deployment^{1–3}. Here we show that unstructured clinical notes from the electronic health record can enable the training of clinical language models, which can be used as all-purpose clinical predictive engines with low-resistance development and deployment. Our approach leverages recent advances in natural language processing^{4,5} to train a large language model for medical language (NYUTron) and subsequently fine-tune it across a wide range of clinical and operational predictive tasks. We evaluated our approach within our health system for five such tasks: 30-day all-cause readmission prediction, in-hospital mortality prediction, comorbidity index prediction, length of stay prediction, and insurance denial prediction. We show that NYUTron has an area under the curve (AUC) of 78.7–94.9%, with an improvement of 5.36–14.7% in the AUC compared with traditional models. We additionally demonstrate the benefits of pretraining with clinical text, the potential for increasing generalizability to different sites through fine-tuning and the full deployment of our system in a prospective, single-arm trial. These results show the potential for using clinical language models in medicine to read alongside physicians and provide guidance at the point of care.

GatorTron models : 345 Million, 3.9 Billion, and 8.9 Billion parameters

npj | digital medicine

www.nature.com/npjdigitalmed

ARTICLE OPEN

A large language model for electronic health records

Xi Yang^{1,2}, Aokun Chen^{1,2}, Nima PourNejatian³, Hoo Chang Shin³, Kaleb E. Smith³, Christopher Parisien³, Colin Compas³, Cheryl Martin³, Anthony B. Costa³, Mona G. Flores^{1,3}, Ying Zhang^{1,4}, Tanja Magoc⁵, Christopher A. Harle^{1,5}, Gloria Lipori^{5,6}, Duane A. Mitchell⁶, William R. Hogan^{1,6}, Elizabeth A. Shenkman^{1,6}, Jiang Bian^{1,2} and Yonghui Wu^{1,2}✉

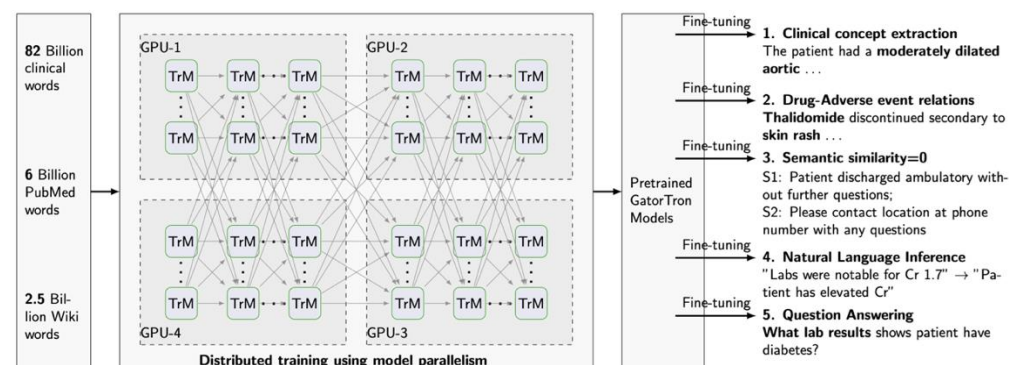
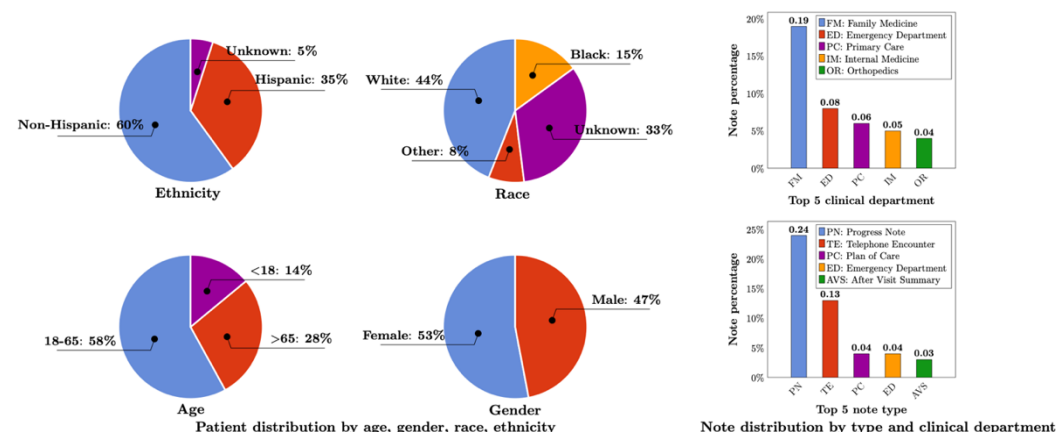
Check for updates

There is an increasing interest in developing artificial intelligence (AI) systems to process and interpret electronic health records (EHRs). Natural language processing (NLP) powered by pretrained language models is the key technology for medical AI systems utilizing clinical narratives. However, there are few clinical language models, the largest of which trained in the clinical domain is comparatively small at 110 million parameters (compared with billions of parameters in the general domain). It is not clear how large clinical language models with billions of parameters can help medical AI systems utilize unstructured EHRs. In this study, we develop from scratch a large clinical language model—GatorTron—using >90 billion words of text (including >82 billion words of de-identified clinical text) and systematically evaluate it on five clinical NLP tasks including clinical concept extraction, medical relation extraction, semantic textual similarity, natural language inference (NLI), and medical question answering (MQA). We examine how (1) scaling up the number of parameters and (2) scaling up the size of the training data could benefit these NLP tasks. GatorTron models scale up the clinical language model from 110 million to 8.9 billion parameters and improve five clinical NLP tasks (e.g., 9.6% and 9.5% improvement in accuracy for NLI and MQA), which can be applied to medical AI systems to improve healthcare delivery. The GatorTron models are publicly available at: https://catalog.ngc.nvidia.com/orgs/nvidia/teams/clara/models/gatortron_og.

npj Digital Medicine (2022)5:194; <https://doi.org/10.1038/s41746-022-00742-2>

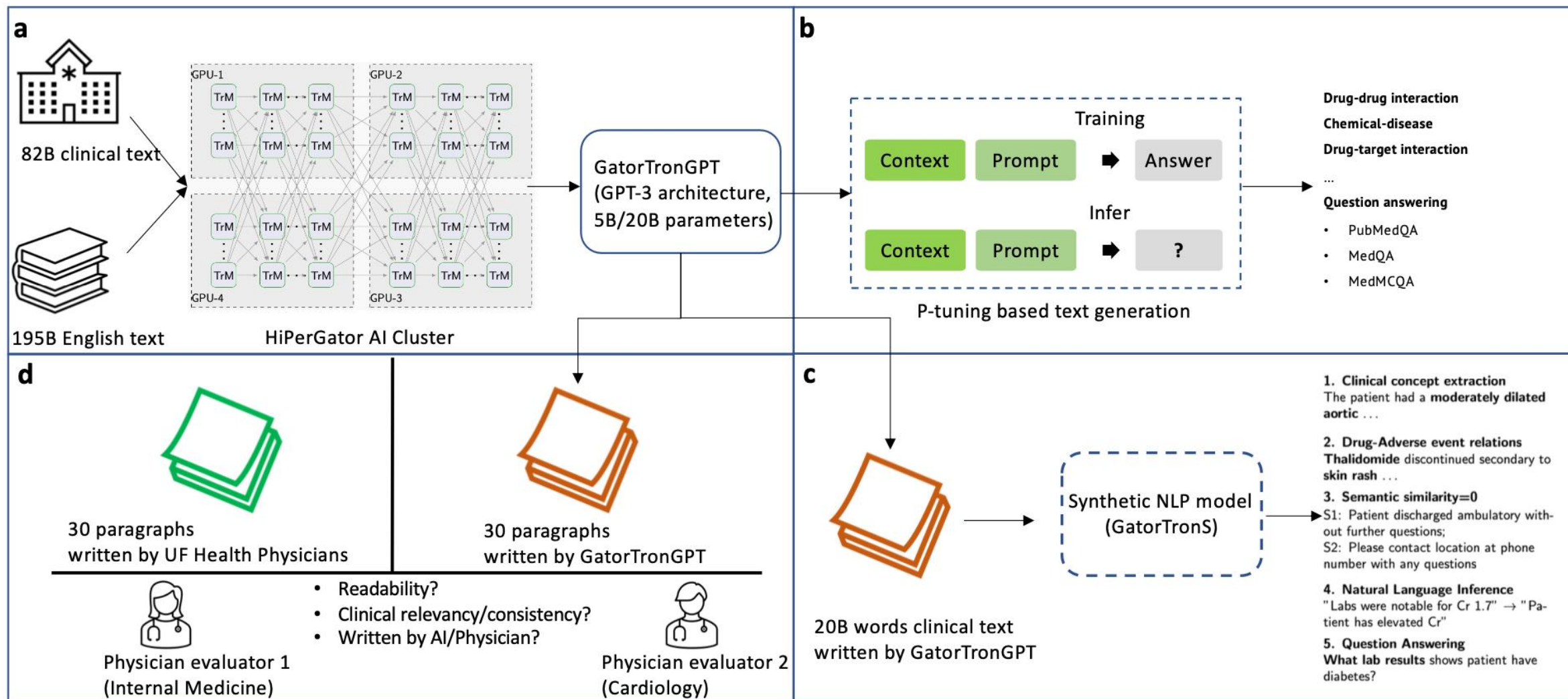


HiperGator-AI with 124 nodes; 992 GPUs; 6 days
Available from: <https://huggingface.co/UFNLP>
Over 3 million downloads



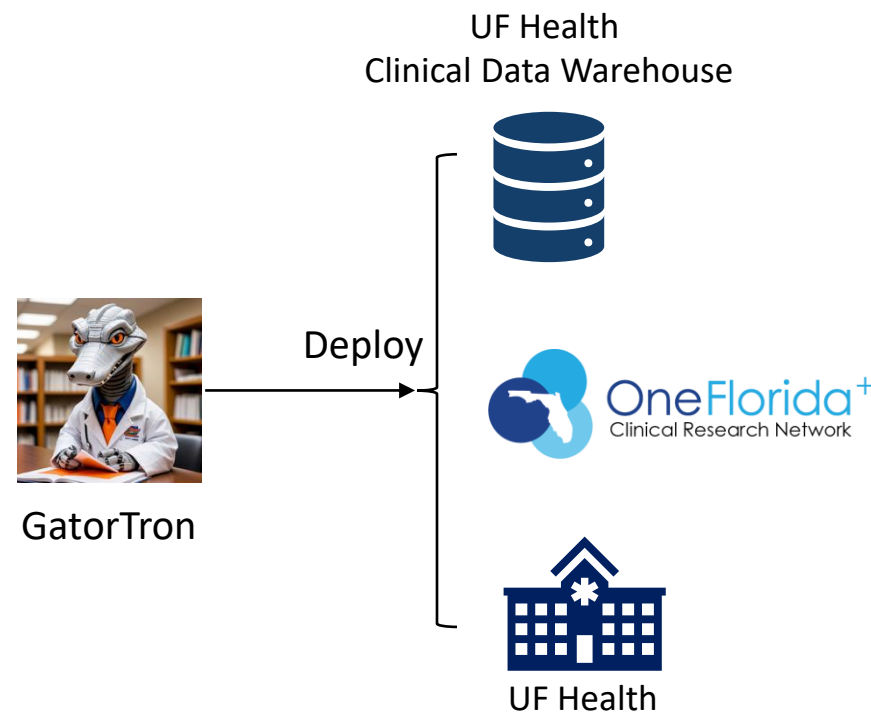
Model	# Layers	# Hidden Size	# Attention Heads	# Parameters
GatorTron-base	24	1024	16	345 million
GatorTron-medium	48	2560	40	3.9 billion
GatorTron-large	56	3584	56	8.9 billion

GatorTronGPT - A generative LLM for EHRs

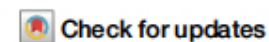


UF CTSI AI/Data Science infrastructure

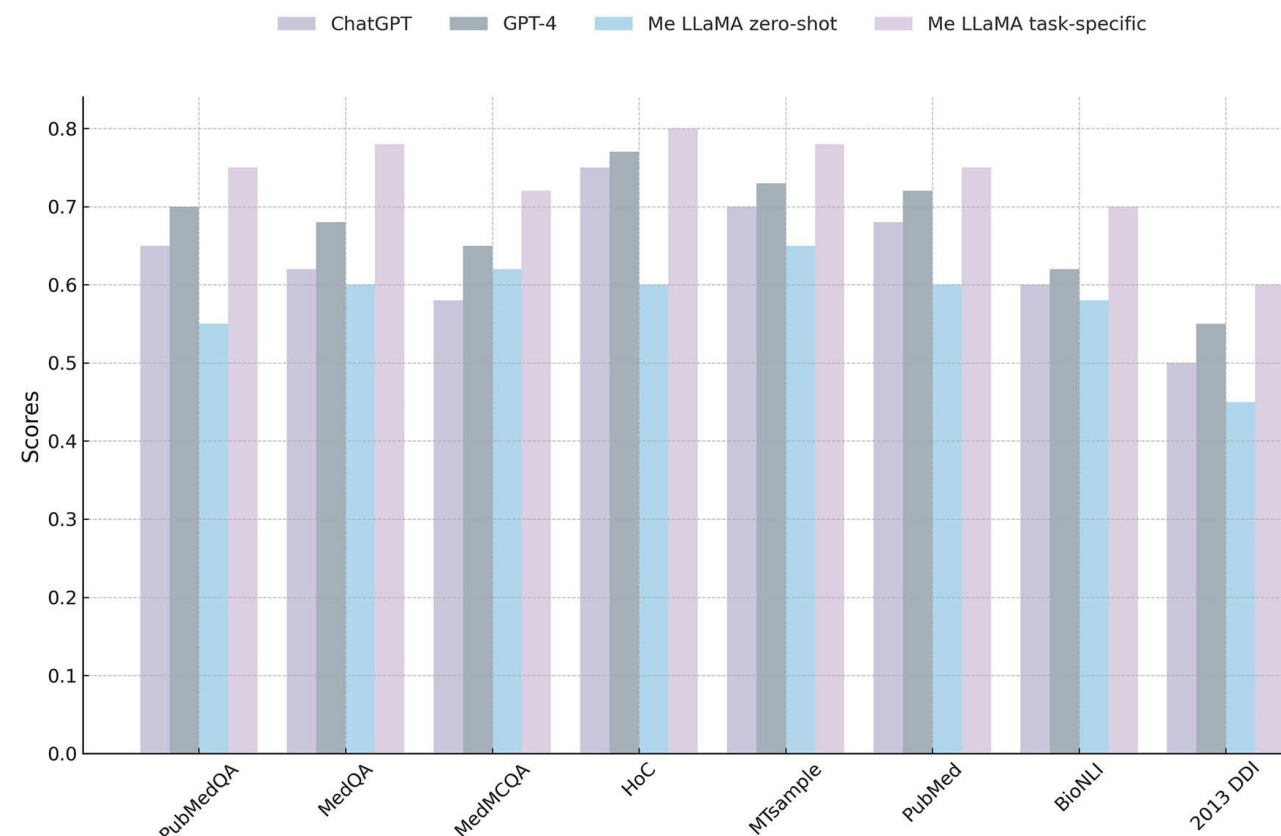
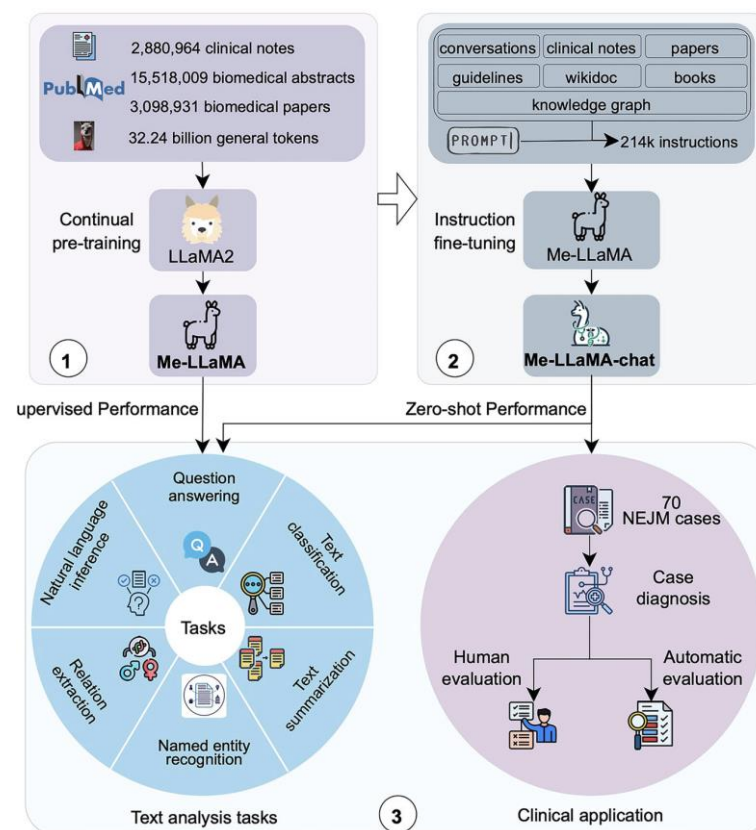
- Healthcare foundation model / generative AI
 - GatorTron - over 2 million downloads
 - GatorTronT5, GatorTron-LLAMA, Multimodal GatorTronV
- NLP infrastructure
 - **DeepDeID, De-identification – Deployed at UF Health IDR since 2020, created a de-identified copy of UF Health notes.**
 - **SODA - Social determinants of health. Deployed and providing service since 2021.**
 - **Quantitative smoking. Deployed and providing service at UF Health IDR since 2021.**
 - **GatorTronCancer, Nodule, tumor, and biomarker extraction for Cancer.**
 - **Opioid use/problematic use.**
 - CIDER- Contextual Infectious Disease Exposure Risk.
 - Cognitive tests, biomarker extraction for Aging
 - Patient fall risk extraction and prediction
 - Delirium symptom extraction and prediction
 - Adverse drug events extraction
 - Transgender and gender non-conforming subjects detection
 - History of suicide attempts/ideation detection
 - Clinical summarization
- Services
 - Computable phenotyping - Accurately identify research cohort
 - Information extraction - fill data gaps
 - Predictive modeling, reasoning – decision support
 - Reduce clinicians' burden – documentation, trial matching, etc.



Medical foundation large language models for comprehensive text analysis and beyond



Qianqian Xie^{1,5}, Qingyu Chen^{1,5}, Aokun Chen^{2,5}, Cheng Peng², Yan Hu³, Fongci Lin¹, Xueqing Peng¹, Jimin Huang¹, Jeffrey Zhang¹, Vipina Keloth¹, Xinyu Zhou¹, Lingfei Qian¹, Huan He¹, Dennis Shung^{1,4}, Lucila Ohno-Machado¹, Yonghui Wu², Hua Xu¹✉ & Jiang Bian²✉



<https://www.nature.com/articles/s41746-025-01533-1>

OpenAI GPT-OSS

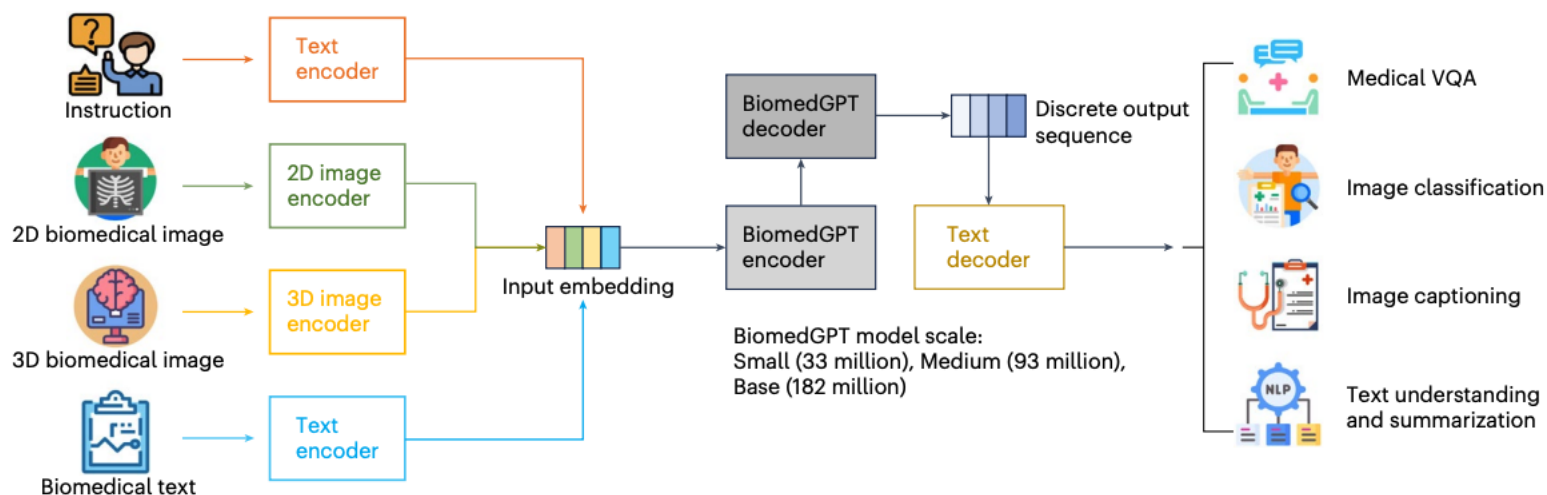
- Open-source Apache 2.0 license, only for text
- Chain-of-thought reasoning
- Mixture-of-Experts (MoE) transformers

Component	120b	20b
MLP	114.71B	19.12B
Attention	0.96B	0.64B
Embed + Unembed	1.16B	1.16B
Active Parameters	5.13B	3.61B
Total Parameters	116.83B	20.91B
Checkpoint Size	60.8GiB	12.8GiB

<https://openai.com/index/introducing-gpt-oss/>

Learn to see

- Multimodal, vision-language model.
 - Make image like text
- Encoder-decoder
 - Unified vision/text token
 - Separate vision/text token: vision encoder + LLM decoder



A generalist vision-language foundation model for diverse biomedical tasks

Med-Flamingo: a Multimodal Medical Few-shot Learner

Michael Moor*

Qian Huang*

Shirley Wu

Michihiro Yasunaga

Yash Dalmia

Jure Leskovec[†]

Department of Computer Science, Stanford University, Stanford,

Cyril Zakka

Department of Cardiothoracic Surgery, Stanford Medicine, Stanf

Eduardo Pontes Reis

Hospital Israelita Albert Einstein, São Paulo, Brazil

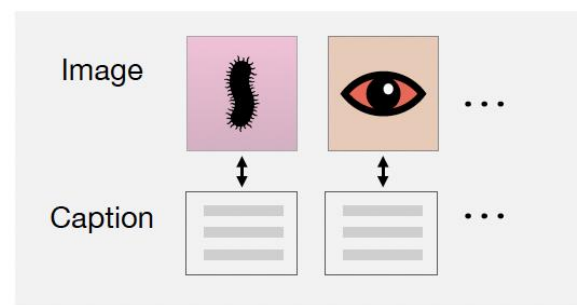
Pranav Rajpurkar

Department of Biomedical Informatics, Harvard Medical School,

CORRESPONDENCE TO: MDMOOR@CS.STANFORD.EDU

1. Multimodal pre-training on medical literature

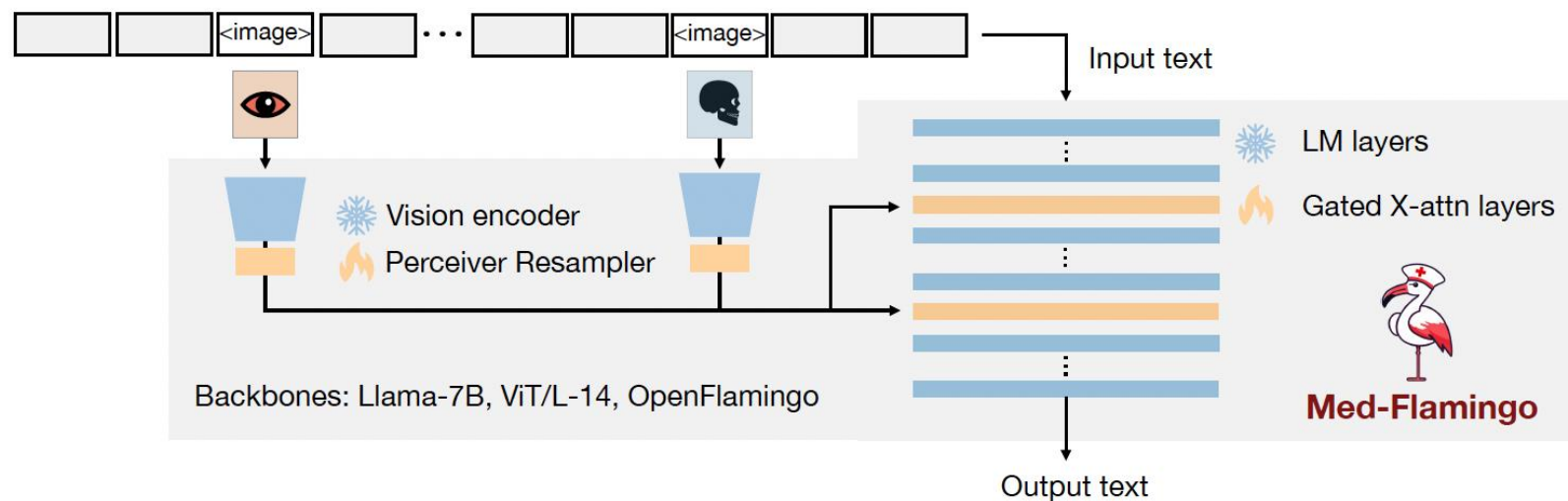
Paired data



Interleaved data



Tokenized data



• OpenFlamingo-9B

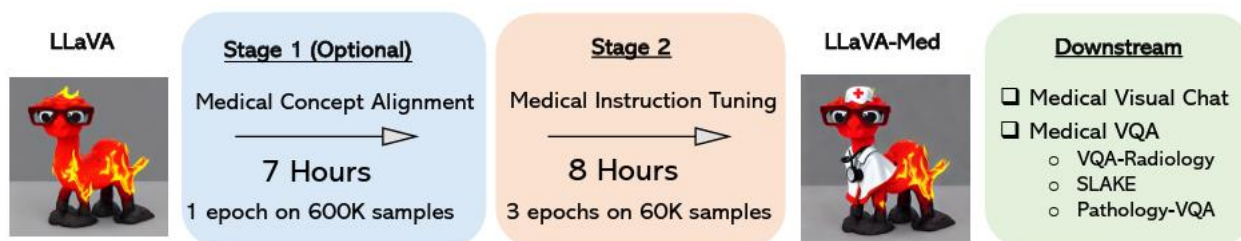
- General VLM
- Language decoder: LLaMA
- Vision encoder: CLIP

LLaVA-Med

- Visual encoder: CLIP
- Language decoder: Vicuna
- Dataset: Image-text pairs

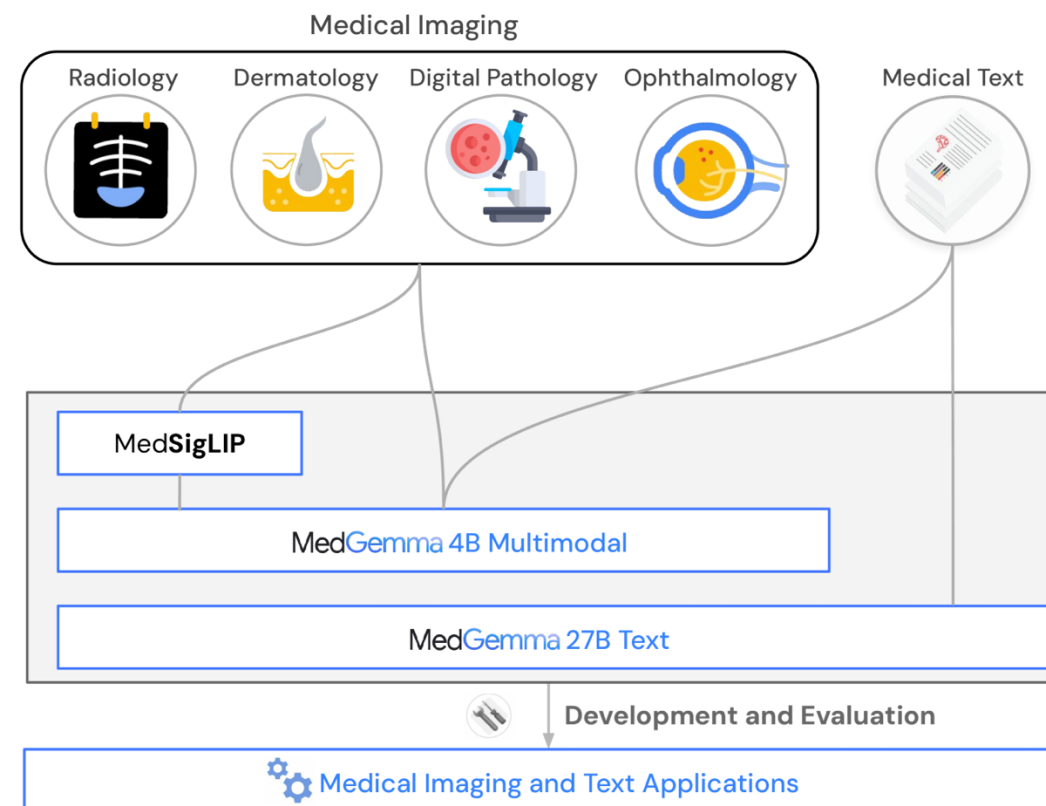
LLaVA-Med: Training a Large Language-and-Vision Assistant for Biomedicine in One Day

Chunyuan Li*, Cliff Wong*, Sheng Zhang*, Naoto Usuyama, Haotian Liu, Jianwei Yang
Tristan Naumann, Hoifung Poon, Jianfeng Gao
Microsoft



Gemma 3 and MedGemma

- Gemma 3: 1b - 27B general vision-language model
 - Vision encoder: SigLIP
 - Language decoder: Gemini
- MedGemma
 - Vision encoder: MedSigLIP



MedGemma Technical Report: <https://arxiv.org/html/2507.05201v2>

Gemma 3 Technical Report: <https://arxiv.org/abs/2503.19786>

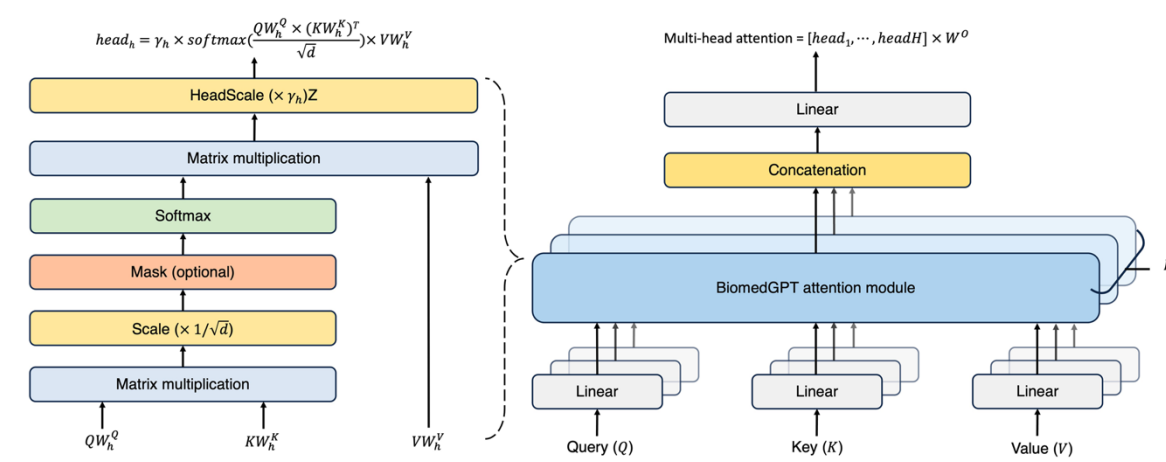
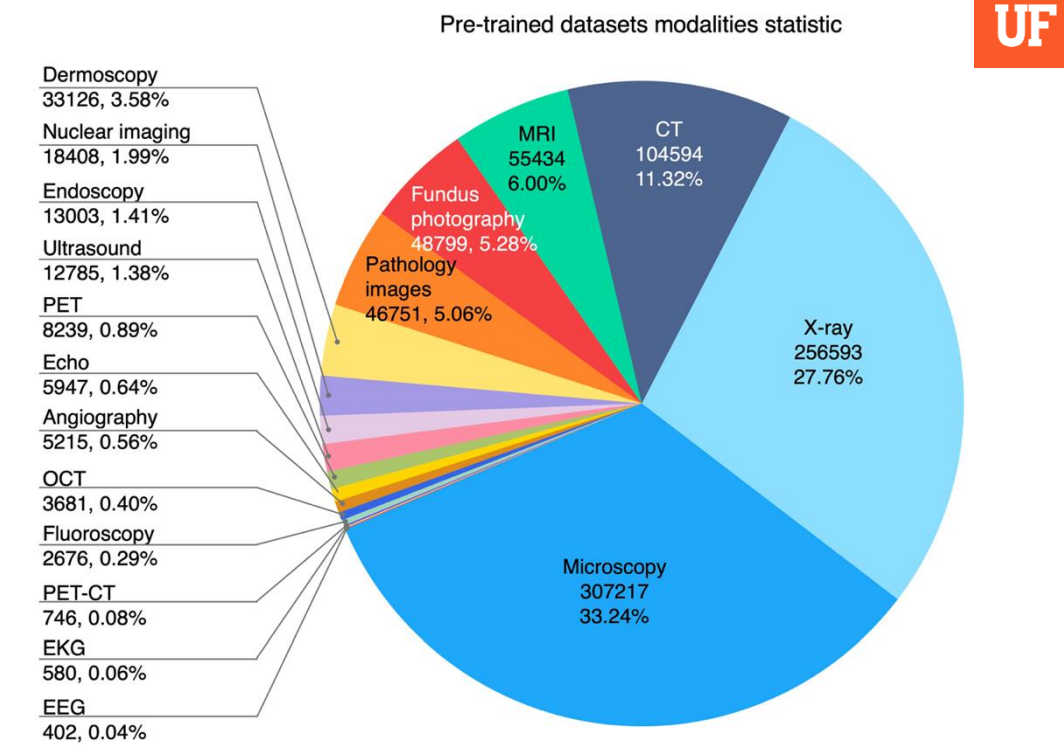
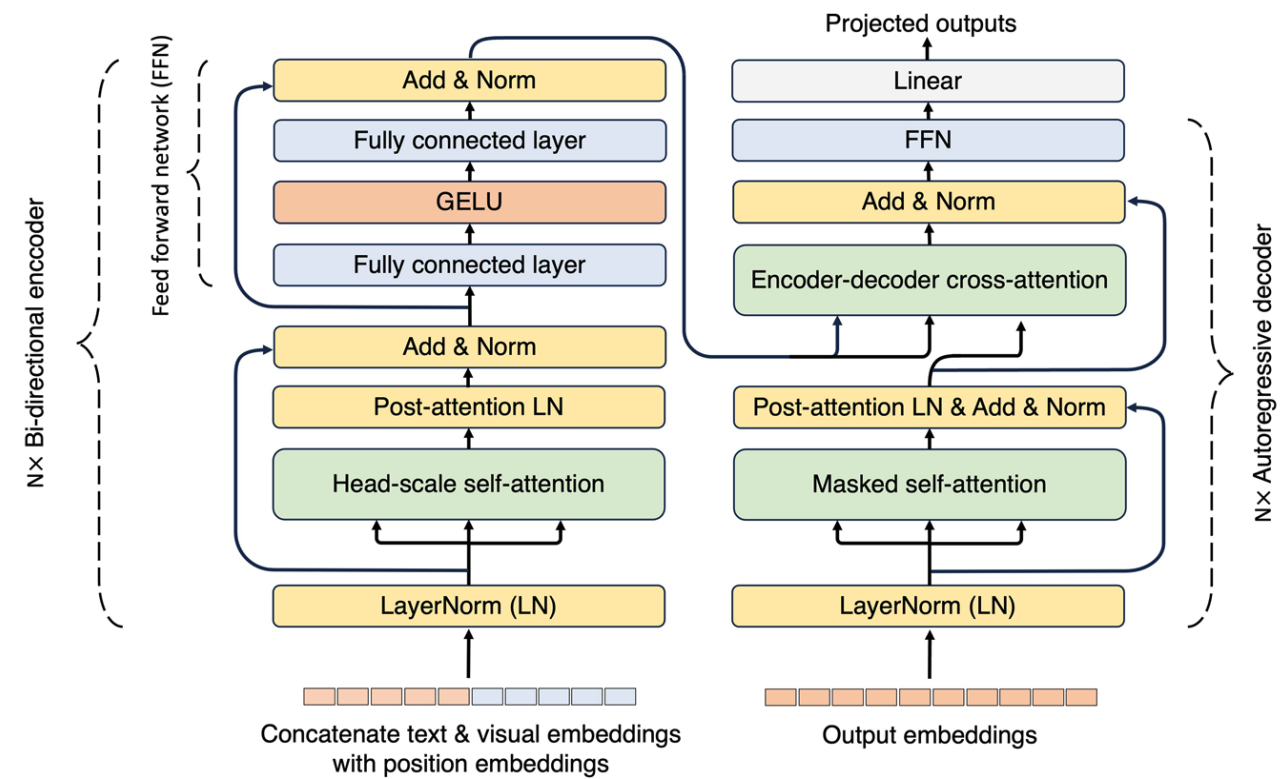
A generalist vision–language foundation model for diverse biomedical tasks

Received: 29 January 2024

Accepted: 10 July 2024

Published online: 7 August 2024

Kai Zhang¹, Rong Zhou¹, Eashan Adhikarla¹, Zhiling Yan¹, Yixin Liu¹, Jun Yu¹, Zhengliang Liu², Xun Chen³, Brian D. Davison¹, Hui Ren⁴, Jing Huang^{5,6}, Chen Chen⁷, Yuyin Zhou⁸, Sunyang Fu⁹, Wei Liu¹⁰, Tianming Liu², Xiang Li⁴, Yong Chen^{5,11,12,13}, Lifang He¹, James Zou^{14,15}, Quanzheng Li⁴, Hongfang Liu⁹ & Lichao Sun¹



Scale-up rule holds



Journal of Biomedical Informatics

Available online 23 October 2025, 104946

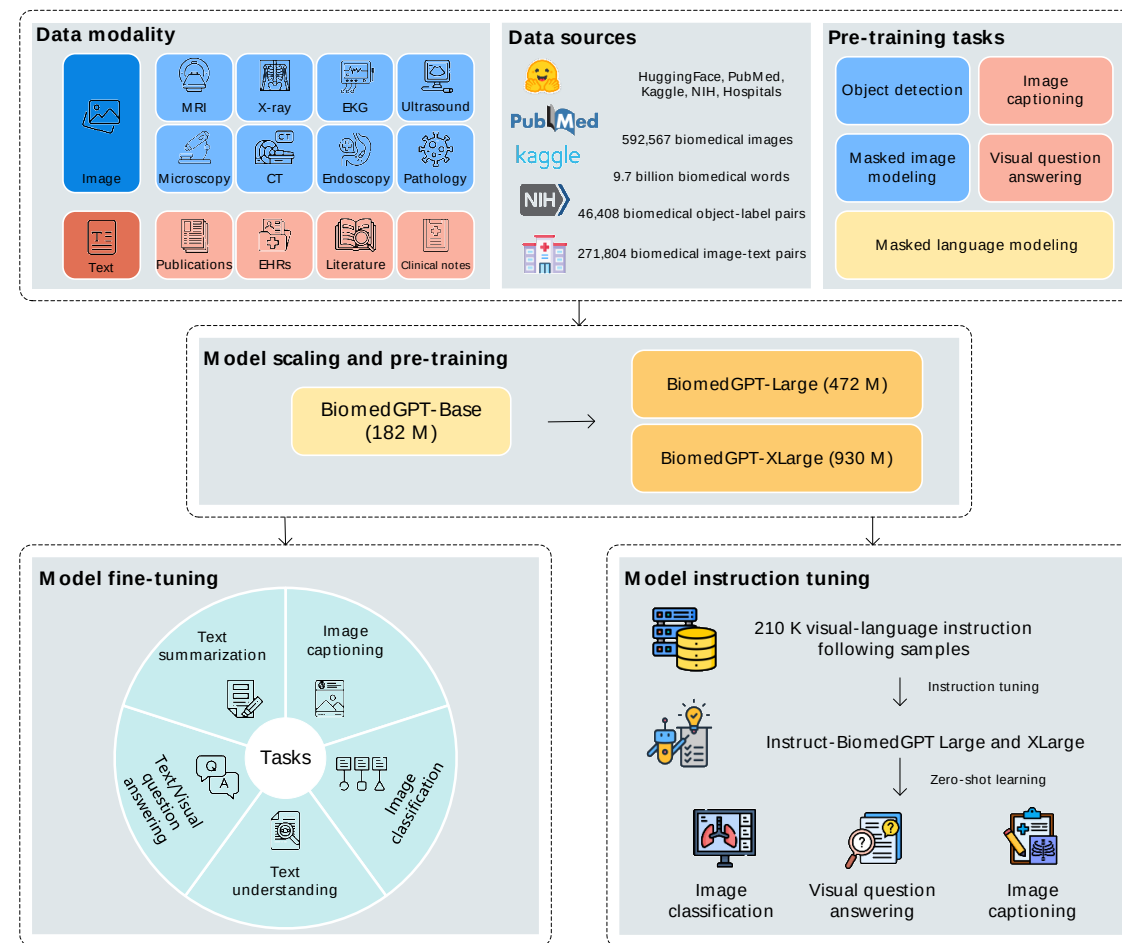
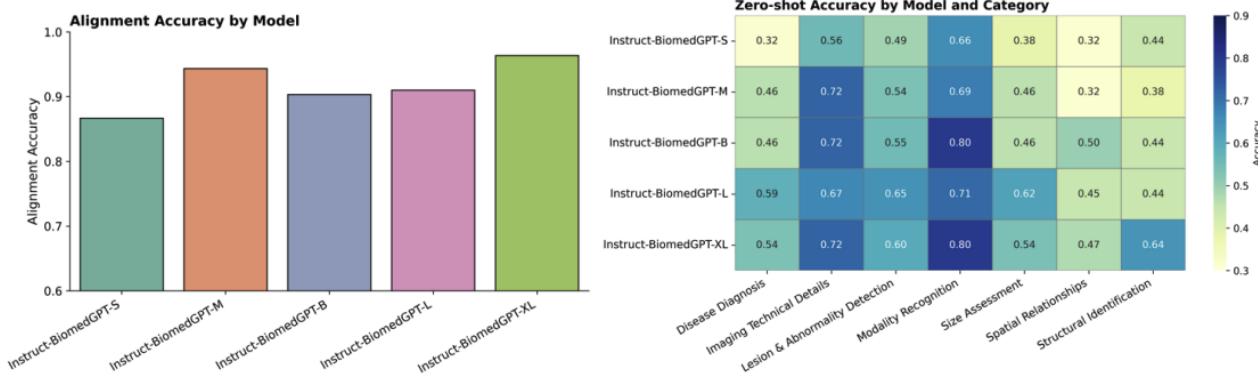
In Press, Journal Pre-proof ? What's this?



Original Research

Scaling up biomedical vision-language models: Fine-tuning, instruction tuning, and multi-modal learning ☆

Cheng Peng ^{a1}, Kai Zhang ^{d1}, Mengxian Lyu ^a, Hongfang Liu ^c, Lichao Sun ^d ,
Yonghui Wu ^{a b}



Learn to think

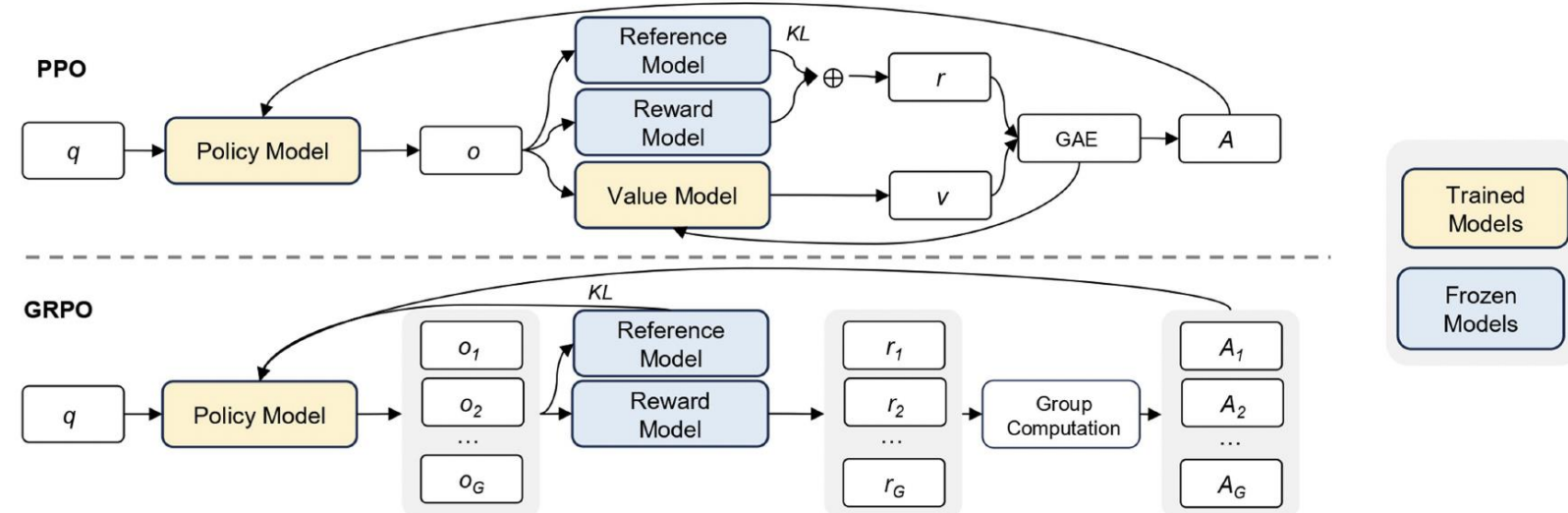
- RL training using GRPO- Group Relative Policy Optimization
- Bypass supervised fine-tuning before RL training

[nature](#) > [articles](#) > [article](#)

Article | [Open access](#) | Published: 17 September 2025

DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning

[Daya Guo](#), [Dejian Yang](#), [Haowei Zhang](#), [Junxiao Song](#), [Peiyi Wang](#), [Qihao Zhu](#), [Runxin Xu](#), [Ruoyu Zhang](#), [Shirong Ma](#), [Xiao Bi](#), [Xiaokang Zhang](#), [Xingkai Yu](#), [Yu Wu](#), [Z. F. Wu](#), [Zhibin Gou](#), [Zhihong Shao](#), [Zhuoshu Li](#), [Ziyi Gao](#), [Aixin Liu](#), [Bing Xue](#), [Bingxuan Wang](#), [Bochao Wu](#), [Bei Feng](#), [Chengda Lu](#), ... [Zhen Zhang](#) [+ Show authors](#)



Qwen3

- Mixture-of-Expert (MoE)
- Integration of the thinking mode and the non-thinking model

Qwen3 Technical Report

Qwen Team



<https://huggingface.co/Qwen>

<https://modelscope.cn/organization/qwen>

<https://github.com/QwenLM/Qwen3>

Abstract

Models	Layers	Heads (Q / KV)	Tie Embedding	Context Length
Qwen3-0.6B	28	16 / 8	Yes	32K
Qwen3-1.7B	28	16 / 8	Yes	32K
Qwen3-4B	36	32 / 8	Yes	128K
Qwen3-8B	36	32 / 8	No	128K
Qwen3-14B	40	40 / 8	No	128K
Qwen3-32B	64	64 / 8	No	128K

Table 2: Model architecture of Qwen3 MoE models.

Models	Layers	Heads (Q / KV)	# Experts (Total / Activated)	Context Length
Qwen3-30B-A3B	48	32 / 4	128 / 8	128K
Qwen3-235B-A22B	94	64 / 4	128 / 8	128K

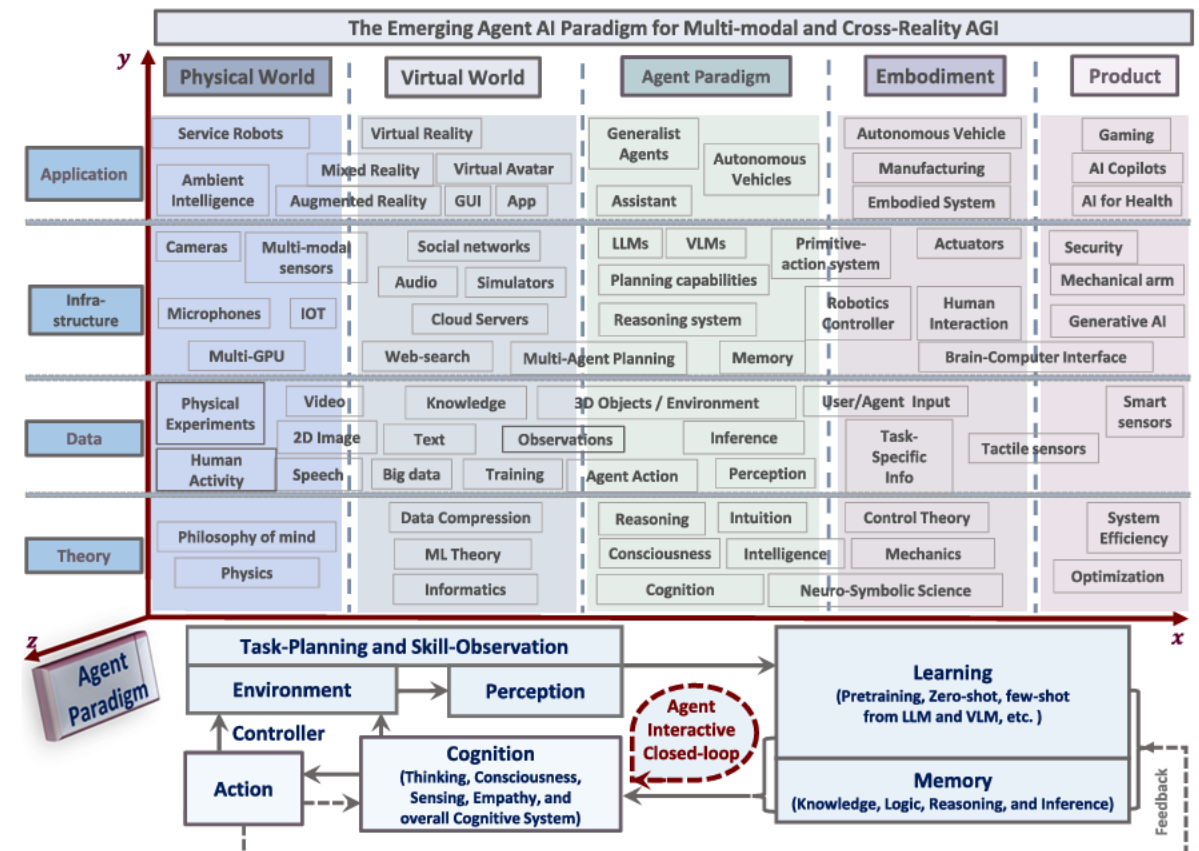
Agentic AI - Collaborative reasoning

- Autonomous systems that can perceive, reason, formulate complex plans, and execute tasks with minimal or no direct human supervision.
- Single agent VS Multiple agents
- Proactive VS reactive
 - LLM: reactive and prompt-driven
 - Agentic AI: proactive and goal-driven
- Reasoning
 - E.g., reasoning through debate

AGENT AI: SURVEYING THE HORIZONS OF MULTIMODAL INTERACTION

Zane Durante^{1†*}, Qiuyuan Huang^{2†*}, Naoki Wake^{2*},
 Ran Gong^{3†}, Jae Sung Park^{4†}, Bidipta Sarkar^{1†}, Rohan Taori^{1†}, Yusuke Noda⁵,
 Demetri Terzopoulos³, Yejin Choi⁴, Katsushi Ikeuchi², Hoi Vo⁵, Li Fei-Fei¹, Jianfeng Gao²

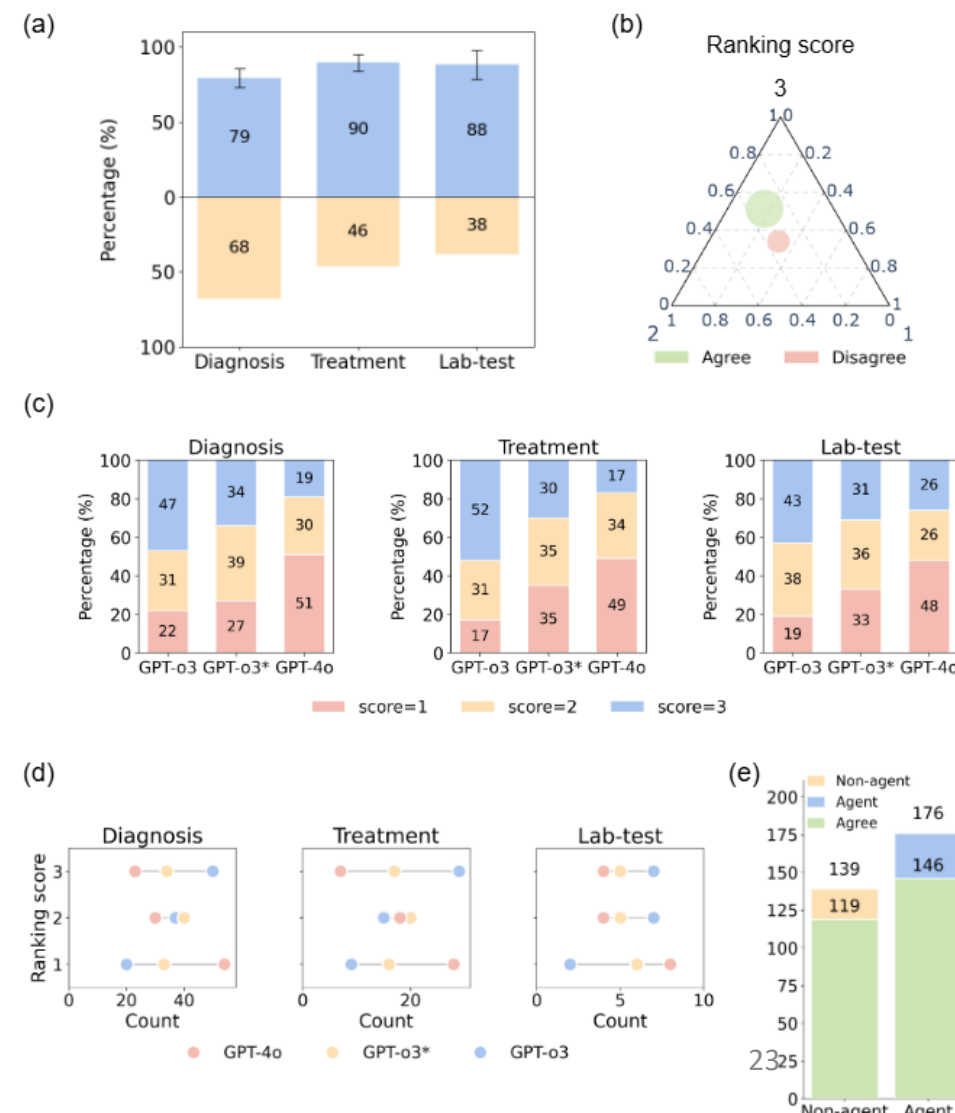
¹Stanford University; ²Microsoft Research, Redmond;
³University of California, Los Angeles; ⁴University of Washington; ⁵Microsoft Gaming



Agentic AI enhances physician-machine consensus in clinical reasoning

- Physicians agree on 84.13% of agentic-reasoning traces (95% CI: [0.8009, 0.8816]), with higher physician ranking compared with non-agree cases ($P=0.0041$).
- Per comparison, agentic AI achieved the highest average physician rankings (up to 34.71% higher than non-agentic AI; $P = 5.8775e - 13$) related to diagnoses, treatments, and lab tests.

Agentic AI enhances physician-machine consensus in clinical reasoning. Yan Z, et. al.



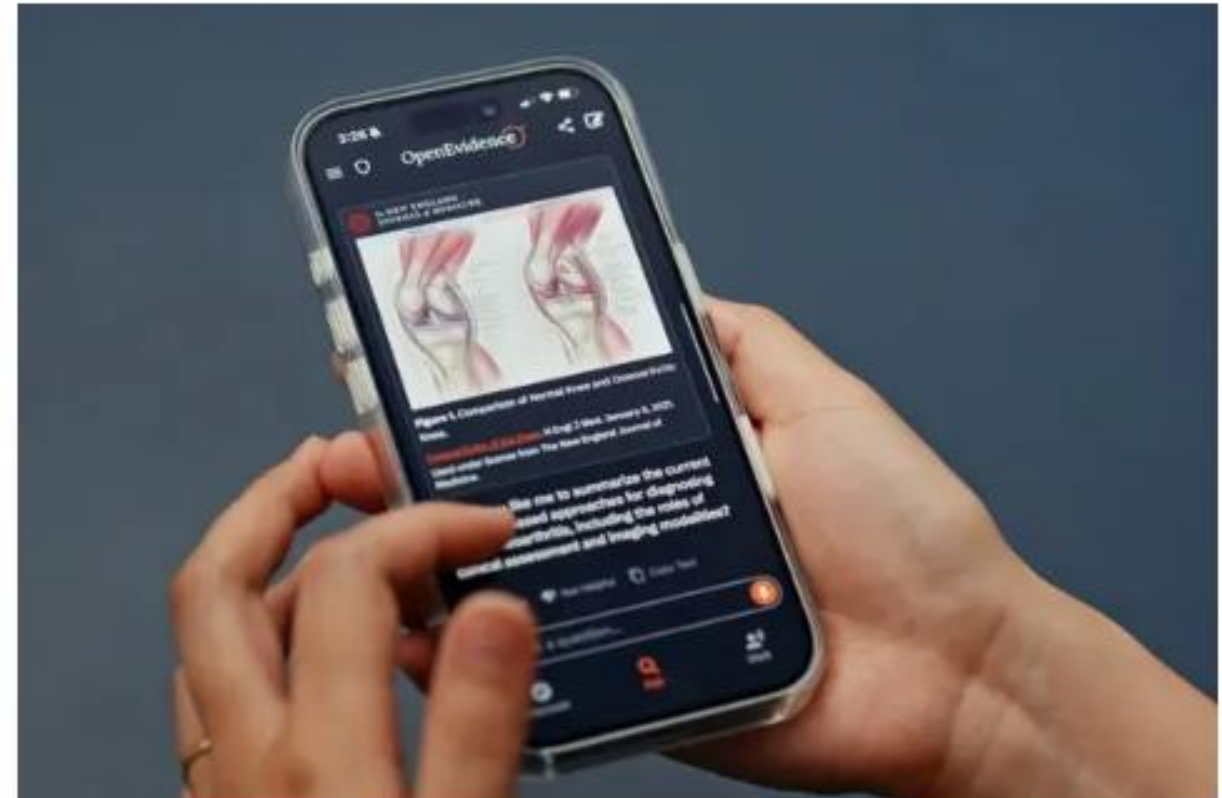
OpenEvidence

OpenEvidence, the Fastest-Growing Application for Physicians in History, Announces \$210 Million Round at \$3.5 Billion Valuation

OpenEvidence also announces the wide release of OpenEvidence DeepConsult™—the first AI agent purpose-built for physicians

OpenEvidence Raises \$200 Million for a ChatGPT for Medicine

The three-year-old artificial intelligence start-up has drawn investor attention, and money, as its use among doctors, nurses and others skyrockets.



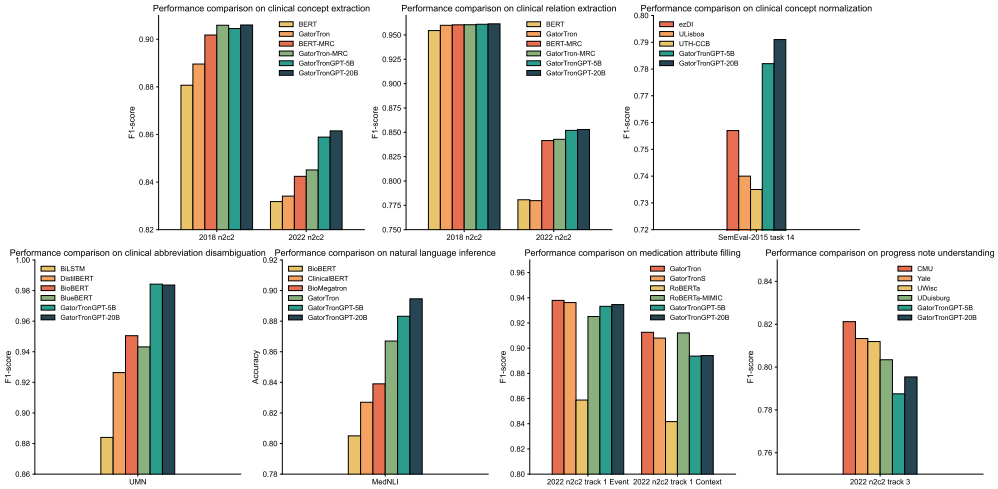
Do we need domain-specific LLMs?

- AGI – artificial general intelligence
- To speak
 - Training from scratch is probably not required.
 - Domain adaptation through post-training is still needed.
- To see and reasoning – yes.
 - Medical images are quite different from general picture
 - Domain knowledge are unique
- Can you trust the evaluation?
 - Data leak
 - Limited samples in the test set
 - Better evaluations are needed.
- **Forgetting** in domain adaptation
 - Overfits to the specific dataset
 - Lost ability to follow human instruction

AGI for NLP



	Input data	Ground truth	Hallucination
1. Nonlogical hallucination: the LLM generated sequences without logic.	Disp: * 60 80 mg syringe * Refills: * 1 * 8. [s2] Acetaminophen [e2] 500 mg Tablet Sig: 1 - 2 Tablets PO [s1] Q6H (every 6 hours) as needed [e1] for pain.	The relation between “Q6H (every 6 hours) as needed” and “Acetaminophen” is “ Frequency-Drug ”.	- Non-responder - Non-responder
	8. docusate sodium 100 mg [s1] Capsule [e1] Sig: One (1) Capsule PO BID (2 times a day). 7. [s2] senna [e2] 8.6 mg Tablet Sig: One (1) Tablet PO BID (2 times a day) as needed for constipation.	The relation between “Capsule” and “senna” is “ No-relation ”.	1: 2: 3: 4: 5: 6: 7: 8: 9: 10: 11: 12: 13: 14:
2. Irrelevant hallucination: the LLM generated readable but irrelevant answers.	Disp: * 30 Tablet (s) * Refills: * 0 * 24. [s2] clopidogrel [e2] 75 mg Tablet Sig: [s1] One (1) [e1] Tablet PO DAILY (Daily)	The relation between “One (1)” and “clopidogrel” is “ Dosage-Drug ”.	1. The drug clopidogrel is a non-steroidal anti-platelet drug that is used to prevent platelet aggregation and reduce the risk of thrombosis in patients with acute coronary syndromes. It is also used to prevent thrombosis in patients with non-cardiac surgery. 2. Clopidogrel is a prodrug that is converted to its active form by the enzyme CYP2C19.
	Medications on Admission: atenolol 25 mg daily aspirin 81 mg daily lipitor 10 mg QOD (every other am) prednisone 10 mg daily tamsulosin SR 0.4 mg evening multivitamin 1 tab daily fish oil capsule 1000 mg twice a day [s2] systane lubricant eye [e2] drops 1 gtt [s1] TID [e1].	The relation between “TID” and “systane lubricant eye” is “ Frequency-Drug ”.	I have been using nonpreserved systane eye drops for a few years.
3. Interpretable hallucination: the LLM generated a different way of the answer that is interpretable.	OXYCODONE - 20 mg Tablet Sustained Release 12 hr - 3 (Three) Tablet (s) by mouth every morning (60 mg), 1 tablet every 2 pm (20 mg) and 3 tablets every evening (60 mg) [s2] PANTOPRAZOLE [e2] [PROTONIX] - 40 mg Tablet , Delayed Release (E.C.) - [s1] 1 [e1] Tablet (s) by mouth once day	The relation between “1” and “PANTOPRAZOLE” is “ Dosage-Drug ”.	The relation type between the Drug entity “PANTOPRAZOLE” and Dosage entity “1” is “has_dosage”.
	Disp: * 30 Tablet (s) * Refills: * 2 * 3. [s2] Fluticasone - Salmeterol [e2] 250 - 50 mcg / Dose Disk with Device Sig: One (1) [s1] Disk with Device [e1] Inhalation Hospital 1 (2 times a day).	The relation between “Disk with Device” and “Fluticasone - Salmeterol” is “ Form-Drug ”	The relation type between the Drug entity “Fluticasone - Salmeterol” and Form entity “Disk with Device” is “is_a”.



1. Clinical Concept Extraction						
Datasets		2018 n2c2			2022 n2c2	
Model	Precision	Recall	F1	Precision	Recall	F1
BERT	0.8887	0.8728	0.8807	0.8160	0.8483	0.8318
GatorTron	0.8759	0.9038	0.8896	0.8181	0.8508	0.8341
BERT-MRC	0.9159	0.8942	0.9018	0.8496	0.8353	0.8424
GatorTron-MRC	0.9199	0.9012	0.9059	0.8521	0.8396	0.8451
GatorTronGPT-5B	0.9168	0.9057	0.9045	0.8598	0.8415	0.8589
GatorTronGPT-20B	0.9189	0.9081	0.9060	0.8615	0.8424	0.8615
2. Clinical Relation Extraction						
Datasets		2018n2c2			2022 n2c2	
Model	Precision	Recall	F1	Precision	Recall	F1
BERT	0.9598	0.9438	0.9545	0.7998	0.7620	0.7807
GatorTron	0.9719	0.9482	0.9599	0.7970	0.7631	0.7798
BERT-MRC	0.9722	0.9489	0.9604	0.8432	0.8371	0.8415
GatorTron-MRC	0.9724	0.9488	0.9605	0.8445	0.8390	0.8428
GatorTronGPT-5B	0.9734	0.9492	0.9610	0.8551	0.8472	0.8520
GatorTronGPT-20B	0.9746	0.9506	0.9614	0.8560	0.8491	0.8529
3. End-to-end Clinical Concept Extraction and Normalization						
Dataset		SemEval-2015 task 14				
Model	Strict Precision	Strict Recall	Strict F1	Relax Precision	Relax Recall	Relax F1
ezDI (best in challenge)	0.783	0.732	0.757	0.815	0.761	0.787
ULisboa	0.779	0.705	0.740	0.806	0.729	0.765
UTH-CCB	0.778	0.696	0.735	0.797	0.714	0.753
GatorTronGPT-5B	0.810	0.751	0.782	0.833	0.769	0.795
GatorTronGPT-20B	0.812	0.772	0.791	0.839	0.785	0.813
4. Clinical Abbreviation Disambiguation						
Dataset	UMN abbreviation			5. Natural Language Inference		
Model	Precision	Recall	F1	Model	MedNLI Accuracy	
BiLSTM	0.8810	0.8960	0.8840	BioBERT	0.8050	
DistilBERT	0.9254	0.9358	0.9263	ClinicalBERT	0.8270	
BioBERT	0.9479	0.9597	0.9505	BioMegatron	0.8390	
BlueBERT	0.9430	0.9518	0.9432	GatorTron	0.8670	
GatorTronGPT-5B	0.9854	0.9832	0.9842	GatorTronGPT-5B	0.8832	
GatorTronGPT-20B	0.9849	0.9830	0.9836	GatorTronGPT-20B	0.8946	
6. Medication Attribute Filling. Dataset: 2022 n2c2 Track 1						
	Event	Context		7. Progress Note Understanding		
Model	F1	F1		Dataset	2022 n2c2	
GatorTron	0.9379	0.9126		Model	F1	
GatorTronS	0.9362	0.9080		CMU (challenge best)	0.8212	
RoBERTa	0.8588	0.8417		Yale	0.8133	
RoBERTa-MIMIC	0.9251	0.9121		UWisc	0.8119	
GatorTronGPT-5B	0.9332	0.8937		UDuisburg	0.8034	
GatorTronGPT-20B	0.9346	0.8941		GatorTronGPT-5B	0.7875	
				GatorTronGPT-20B	0.7954	

Do we need multimodality?

- Language is all you need – all information from different data modalities can be described in language
- Yes, machines can see subtle traits that human eyes can not see
- Agentic AI
- Phased-array CT
 - 3072x3072 resolution
 - 64 fold increase in resolution
 - Determine nodules as cancer



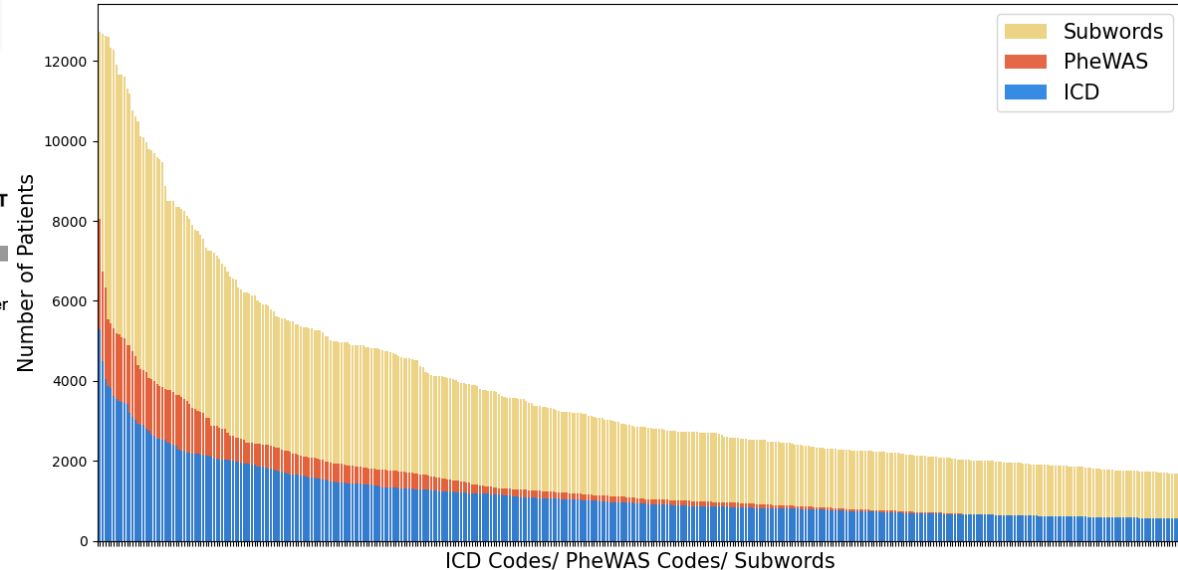
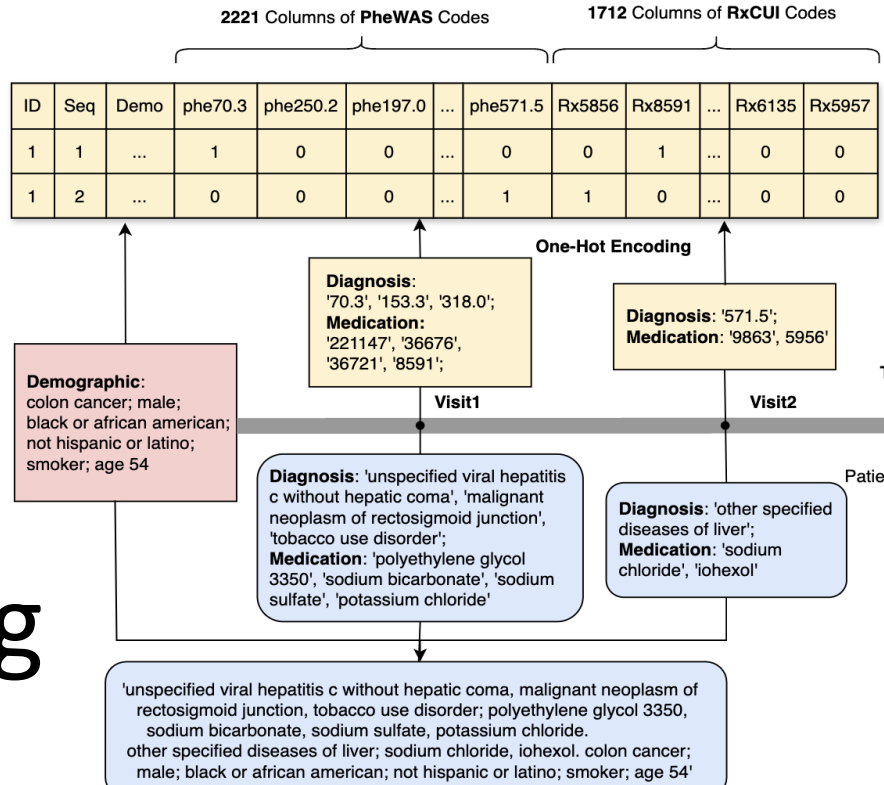
Beyond Rotation: How Phased Array CT Is Redefining AI Medical (A 3-Part Deep-Dive into the Future of Medical Imaging and Intelligent Healthcare)



Li Yuan

COO | Phased Array CT to Empower AI Medical

Clinical Decision Support Through Reasoning



Model	Feature encoding	F1	Precision	Recall	AUC	Specificity	Accuracy
XGB	One-hot	0.290	0.592	0.192	0.825	0.982	0.889
SVMs	One-hot	0.302	0.376	0.252	0.756	0.944	0.863
T-LSTM	Sequence	0.622	0.650	0.596	0.907	0.957	0.914
BERT	Subword	0.643	0.631	0.656	0.890	0.949	0.914
GatorTron-345M	Subword	0.692	0.716	0.669	0.930	0.965	0.929
GatorTron-3.9B	Subword	0.699	0.684	0.715	0.896	0.956	0.927



Ziyi Chen, MS

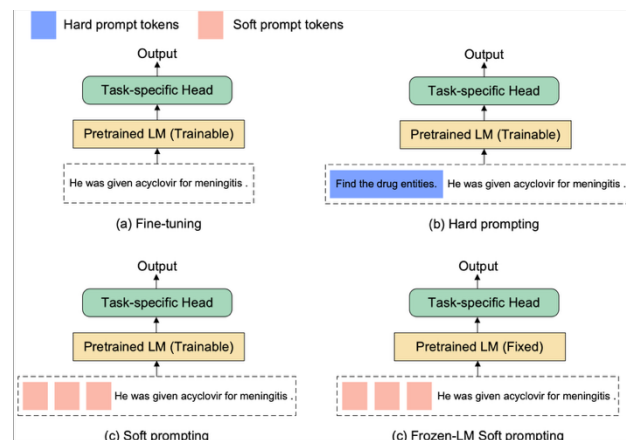
Prompt is not mature

- Limited control power – compared with fine-tuning
- Not robust – works for some instances, doesn't work for other instances
- Not controllable – no systematic way to develop good prompts
- Few-shot/zero-shot good enough for real-world applications



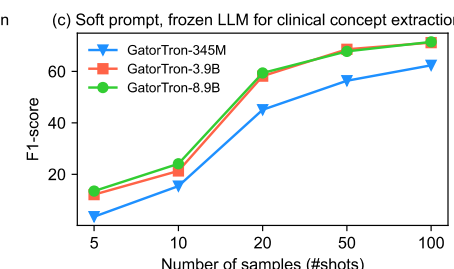
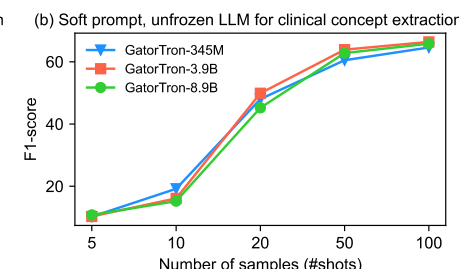
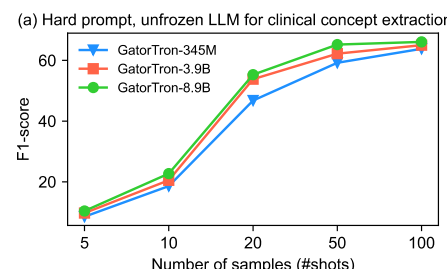
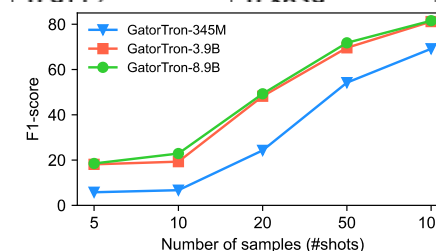
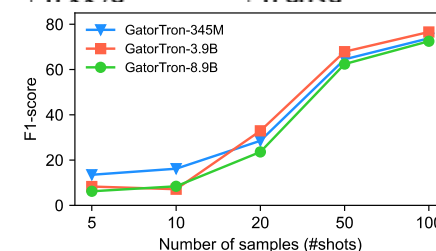
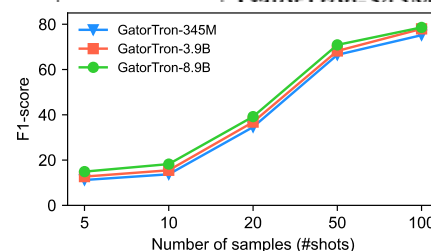
Cheng Peng, PhD

Prompt tuning – make machine learn robust, controllable prompts



- Machines learn soft prompts better than humans
- Frozen LLMs has better few-shot learning ability and transfer learning ability
- Frozen LLMs require large models

Corpus	Models	Training strategies			
		No prompt; Unfrozen LLM	Hard prompt; Unfrozen LLM	Soft prompt; Unfrozen LLM	Soft prompt; Frozen LLM
2018 n2c2	BERT	0.8807	0.9018	0.9095	0.8598
	BERT-MIMIC	0.8853	0.9031	0.9080	0.8676
	RoBERTa	0.8812	0.9016	0.9076	0.8490
	RoBERTa-MIMIC	0.8907	0.9020	0.9081	0.8621
	GatorTron-345M	0.8870	0.9050	0.9112	0.8650



It's useful matters, not become perfect

- Useful means “better than current options”
 - AI models criticized by researchers can be life-saver.
- Screen reading – “The picture shows a dog sitting on the grass.”
- Auto guidance system

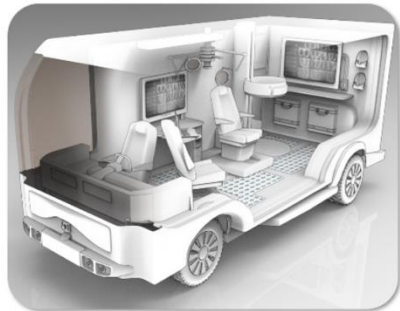


<https://www.youtube.com/watch?v=2lqG43CYQpw>

AI for rural health

Study Highlights

- Multimodal multi-medical task adaptive guidance system that can be deployed on mobile health units
- Instruct rural healthcare workers to perform necessary clinical procedures such as ultrasound and CT scans to improve early detection and reduce morbidity and mortality



PARADIGM Solutions

TA1: Decentralized Approaches to Hospital Level Care (field demos demonstrate clinical effectiveness and sustainability)

TA2: Care Delivery Platform Integration (creation of the scalable / multifunctional care delivery platform)

TA3: Medical IoT Platform (harmonize diverse medical device data with diverse EHR systems)

TA4: Rugged CT Scanner (novel CT tech to bring advanced imaging directly into communities)

TA5: Intelligent Task Guidance (enable healthcare workers to perform tasks beyond their usual training)

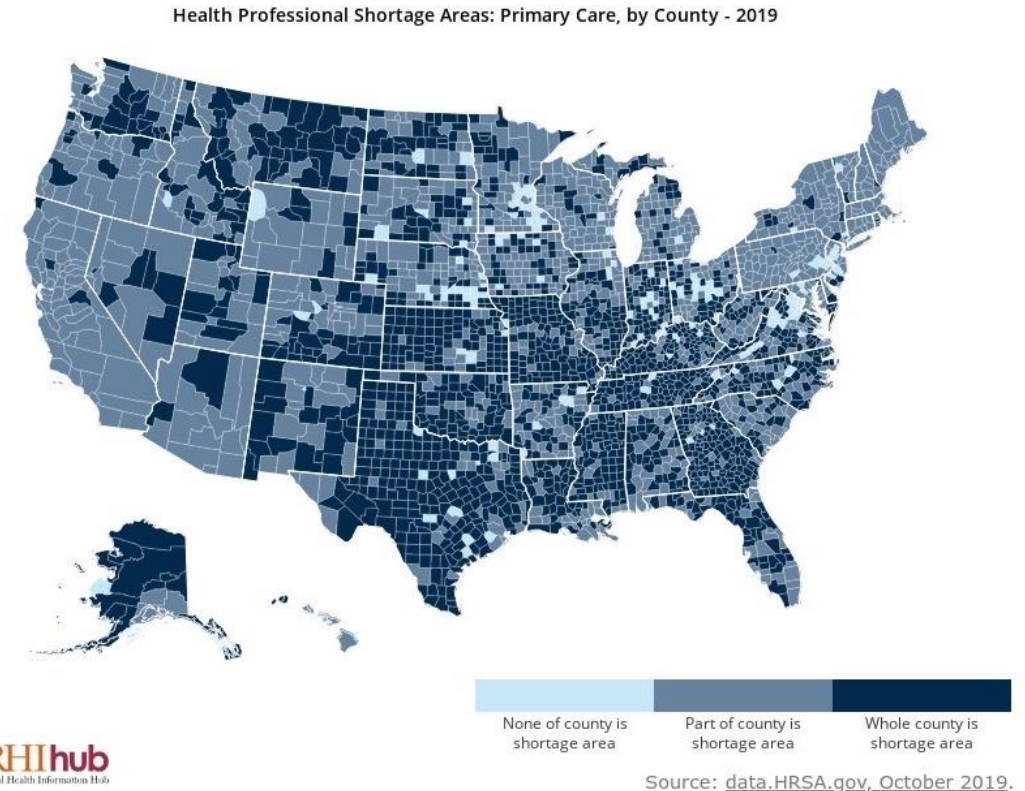
Distributed
Care Platform

Advanced
Imaging

Workforce
Multiplier



GatorTron



ARPAH - Multimodal Multi-Medical Task Adaptive Guidance System (Multi-TAGs).
(Wu, MPI, 11/14/2024 - 11/13/2029)

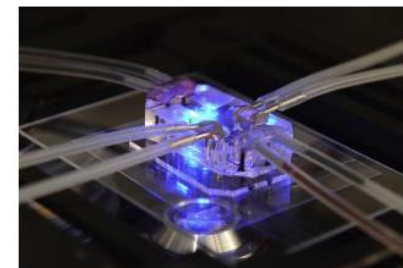
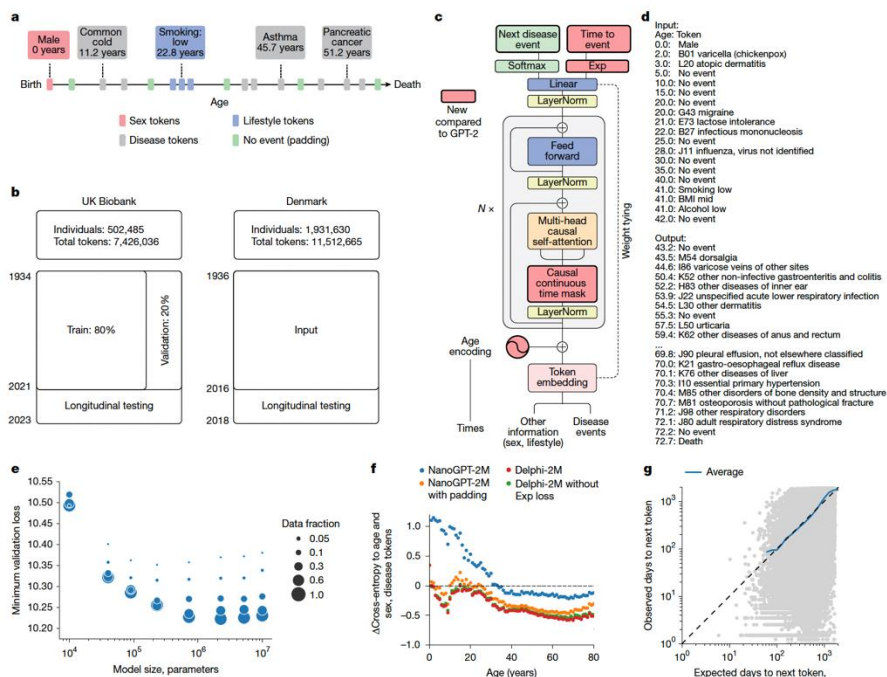
Simulation, not one-time classification

Article | [Open access](#) | Published: 17 September 2025

Learning the natural history of human disease with generative transformers

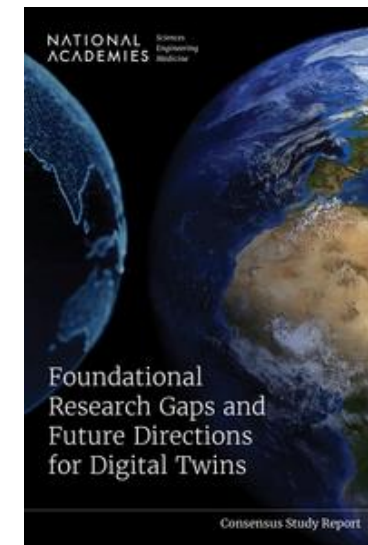
[Artem Shmatko](#), [Alexander Wolfgang Jung](#), [Kumar Gaurav](#), [Søren Brunak](#), [Laust Hvas Mortensen](#), [Ewan Birney](#) , [Tom Fitzgerald](#)  & [Moritz Gerstung](#) 

[Nature](#) (2025) | [Cite this article](#)



Combining microfabrication techniques with modern tissue engineering, the lung-on-a-chip, designed by the Wyss Institute at Harvard

NIH initiative to expand innovative, human-based science while reducing animal use



WestWorld - Caleb-279



Black Mirror, 998/1000 simulations

